

Question 1 Part 1

Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

# Step 1: Load dataset
df = pd.read_csv("Titanic-Dataset.csv")

# Step 2: Show structure and summary
print("== Dataset Info ==")
df.info()
print("\n== Statistical Summary ==")
print(df.describe())
```

Output

```
== Dataset Info ==
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
...
```

Analysis

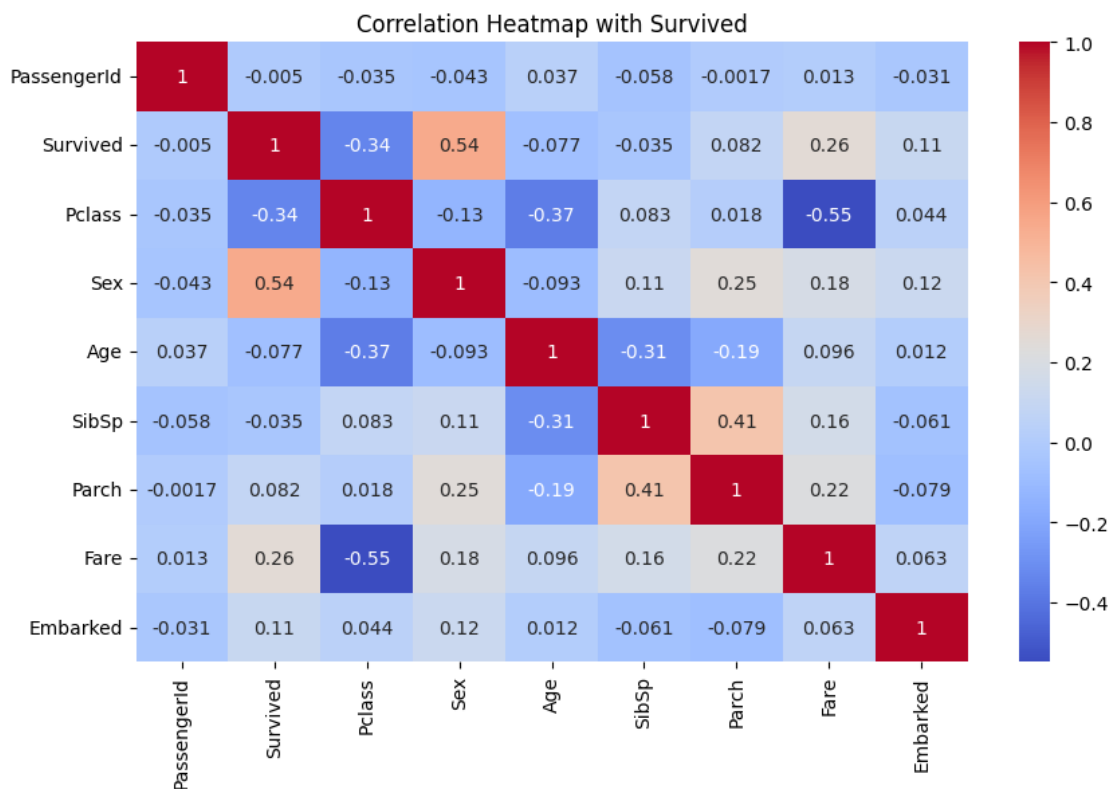
The code loads the Titanic dataset and displays its structure and summary. `df.info()` shows there are 891 entries, with some missing values in columns like **Age** and **Cabin**. `df.describe()` provides statistics such as average age (29.7), average fare (32.2), and survival rate (~38%). It also reveals most passengers traveled alone, helping to understand the dataset's basic characteristics.

```
# Step 3 5 : Encoding, Cleaning, Correlation Heatmap
df_encoded = df.copy()
df_encoded.loc[df_encoded['Sex'] == 'male', 'Sex'] = 0
df_encoded.loc[df_encoded['Sex'] == 'female', 'Sex'] = 1
df_encoded.loc[df_encoded['Embarked'] == 'S', 'Embarked'] = 0
df_encoded.loc[df_encoded['Embarked'] == 'C', 'Embarked'] = 1
df_encoded.loc[df_encoded['Embarked'] == 'Q', 'Embarked'] = 2

df_corr = df_encoded.drop(columns=['Name', 'Ticket', 'Cabin'])
df_corr['Sex'] = df_corr['Sex'].astype(float)
df_corr['Embarked'] = df_corr['Embarked'].astype(float)

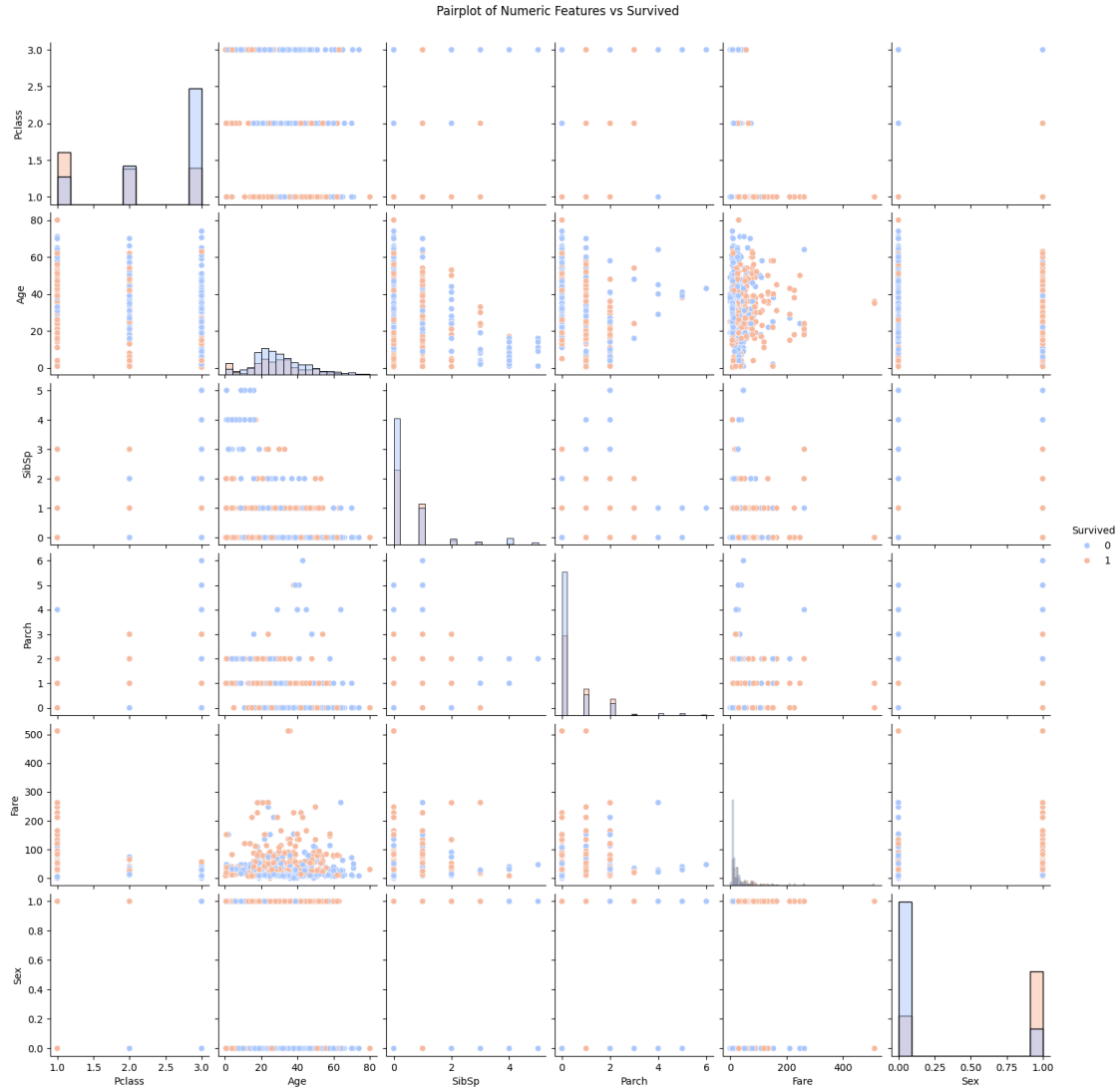
plt.figure(figsize=(10,6))
```

```
sns.heatmap(df_corr.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap with Survived")
plt.show()
```



This section encodes categorical columns (**Sex**, **Embarked**) to numeric form, drops non-numeric columns, and generates a correlation heatmap to examine relationships with the survival outcome. We see **Sex** and **Fare** are among the more influential variables.

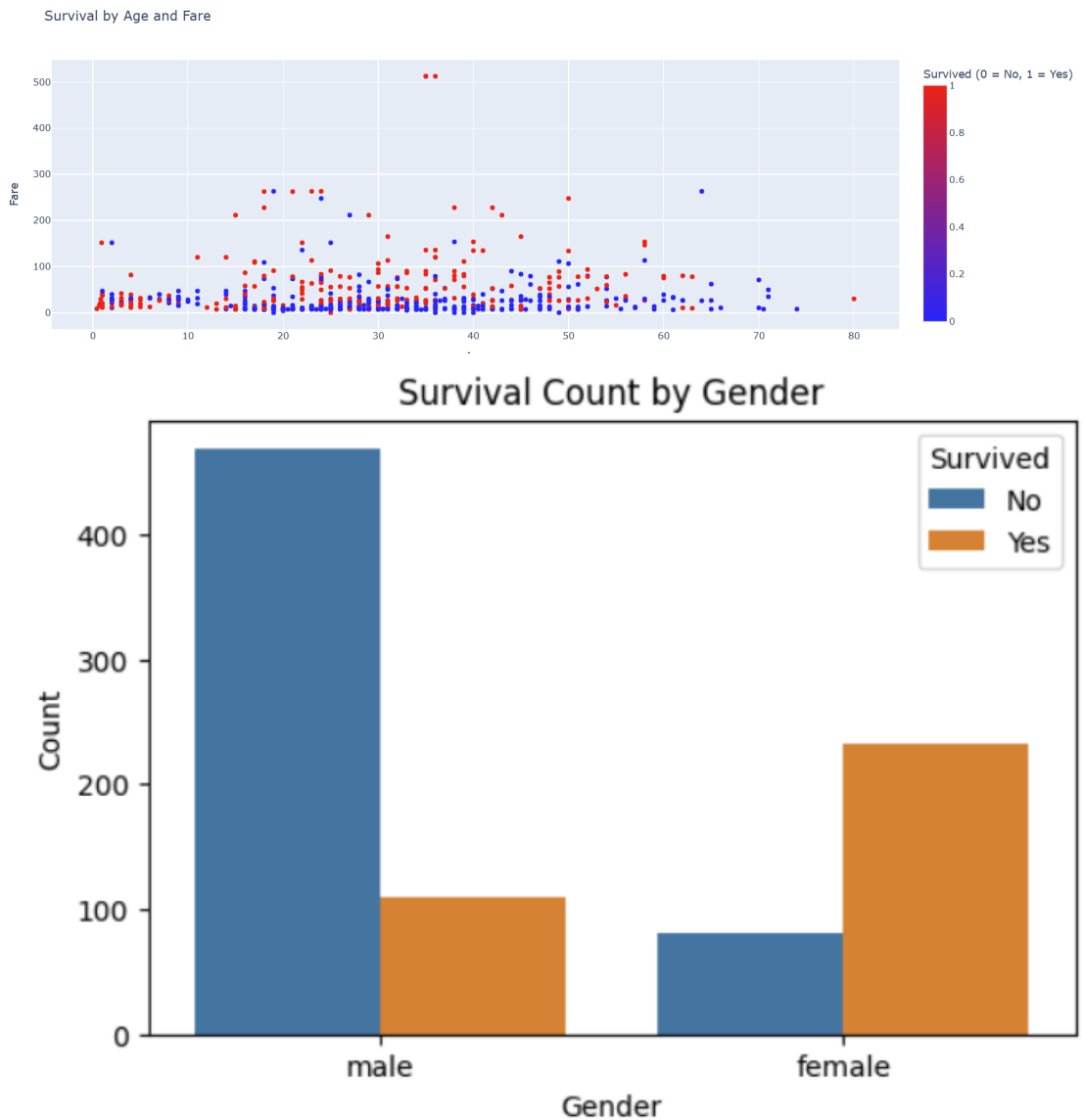
```
# Step 6: Pairplot for numeric features
plot_cols = ['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare', 'Sex']
df_plot = df_encoded[plot_cols].dropna()
sns.pairplot(df_plot, hue='Survived', diag_kind='hist', palette='coolwarm')
plt.suptitle('Pairplot of Numeric Features vs Survived', y=1.02)
plt.show()
```



This pairplot compares survival across numeric features. However, some plots suffer from overlapping points. Additionally, since `Pclass` is categorical, alternative visualizations like grouped bar charts may better illustrate patterns.

```
# Step 7 & 8: Age vs Fare (scatter) and Survival by Gender (countplot)
df_scatter = df[['Age', 'Fare', 'Survived']].dropna()
fig = px.scatter(
    df_scatter,
    x='Age',
    y='Fare',
    color='Survived',
    title='Survival-by-Age-and-Fare',
    labels={'Survived': 'Survived (0=No, 1=Yes)'},
    color_continuous_scale='Bluered'
)
fig.show()
```

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='Sex', hue='Survived')
plt.title('Survival-Count-by-Gender')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()
```



The scatter plot explores how **Age** and **Fare** relate to survival, while the countplot confirms that females had a significantly higher survival rate. These visualizations emphasize demographic and economic influences on survival outcomes.

```
# Step 9: Family Size Analysis
```

```
df['FamilySize'] = df['SibSp'] + df['Parch'] + 1
```

```
family_survival = df.groupby('FamilySize')['Survived'].mean().reset_index()
```

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(data=family_survival, x='FamilySize', y='Survived', palette='viridis')
```

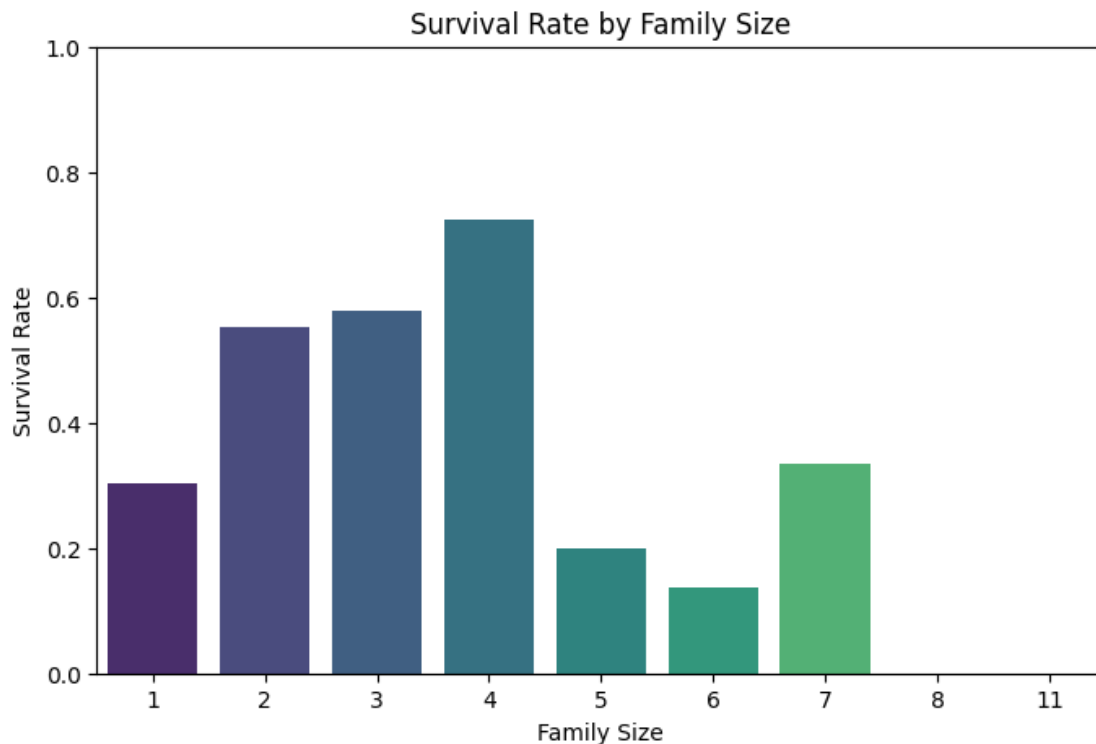
```
plt.title("Survival-Rate-by-Family-Size")
```

```
plt.xlabel("Family-Size")
```

```
plt.ylabel("Survival-Rate")
```

```
plt.ylim(0, 1)
```

```
plt.show()
```



The `FamilySize` column provides a new perspective on social dynamics. Moderate family sizes show better survival rates, while those traveling alone or with large families had lower survival rates—possibly due to social or logistical challenges during evacuation.

```
# Step 10: Alone vs With Family
```

```
df['Alone'] = df['FamilySize'].apply(lambda x: 1 if x == 1 else 0)
```

```
alone_survival = df.groupby('Alone')['Survived'].mean().reset_index()
```

```
alone_survival['Alone'] = alone_survival['Alone'].map({1: 'Alone', 0: 'With-Family'})
```

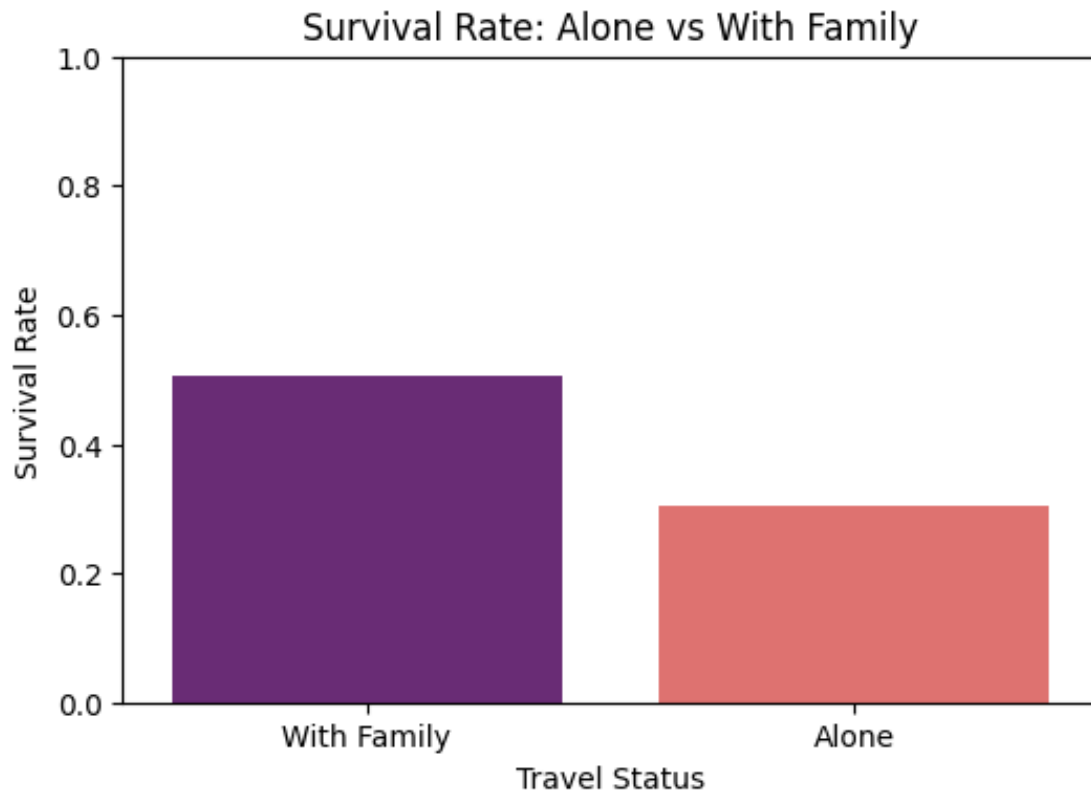
```
plt.figure(figsize=(6, 4))
```

```
sns.barplot(data=alone_survival, x='Alone', y='Survived', palette='magma')
```

```
plt.title("Survival-Rate:-Alone-vs-With-Family")
```

```
plt.xlabel("Travel-Status")
```

```
plt.ylabel("Survival Rate")
plt.ylim(0, 1)
plt.show()
```



This final section introduces the `Alone` column to distinguish passengers traveling solo from those with family. It calculates the survival rate for each group and presents the results in a bar chart.

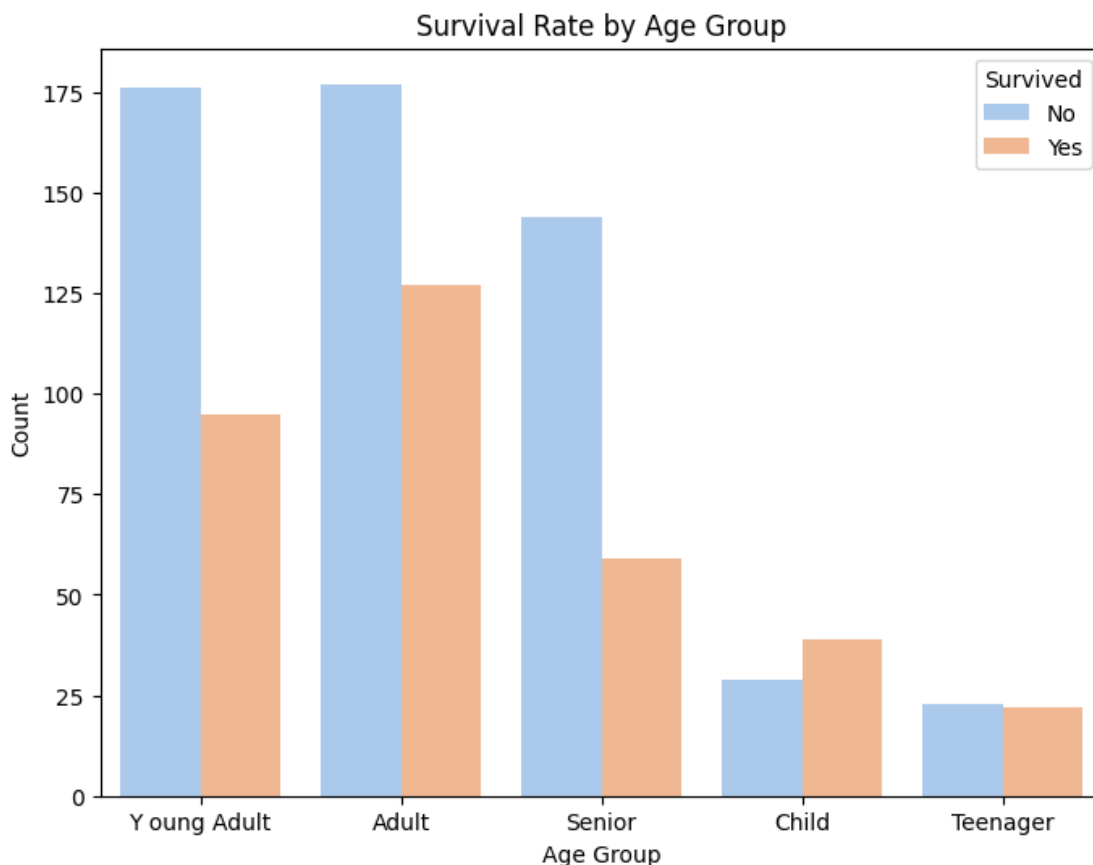
The analysis reveals that passengers who traveled with family had higher survival rates than those who traveled alone. This makes intuitive sense, as group dynamics can lead to coordinated action, emotional support, and protection in emergencies. In contrast, individuals traveling alone may have lacked assistance or priority during evacuation. This insight deepens our understanding of the social factors that influenced survival on the Titanic.

— Age Group Categorization and Survival Analysis —

```
def categorize_age(age):
    if age < 12:
        return 'Child'
    elif 12 <= age < 18:
        return 'Teenager'
    elif 18 <= age < 30:
        return 'Young-Adult'
    elif 30 <= age < 60:
        return 'Adult'
    else:
        return 'Senior'

df_encoded['AgeGroup'] = df_encoded['Age'].apply(categorize_age)
```

```
plt.figure(figsize=(8, 6))
sns.countplot(data=df_encoded, x='AgeGroup', hue='Survived', palette='pastel')
plt.title('Survival Rate by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Count')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()
```



This section focuses on analyzing survival rates by dividing passengers into distinct age groups. The function `categorize_age()` segments individuals based on commonly accepted life stages :

- Child: under 12 years
- Teenager: 12 to under 18
- Young Adult: 18 to under 30
- Adult: 30 to under 60
- Senior: 60 and above

Each passenger's age is passed through this function to generate a new categorical column called `AgeGroup`. This enriched dataset allows for the visual breakdown of survival outcomes across life stages.

A Seaborn countplot is used to show the number of passengers in each age group, split by survival status. The x-axis represents the age categories, while the y-axis shows the total number of passengers in each group. Two colored bars per group display the number of survivors and non-survivors, making comparisons visually intuitive.

This plot is especially helpful for identifying trends such as:

- High survival rates among children due to evacuation priorities.
- Lower survival rates for adults and seniors who may have had less assistance or lower priority.
- Interesting behavioral or cultural dynamics that influenced outcomes across age demographics.

This form of grouping not only enhances data interpretation but also allows for more tailored modeling or subgroup analysis in further studies.